

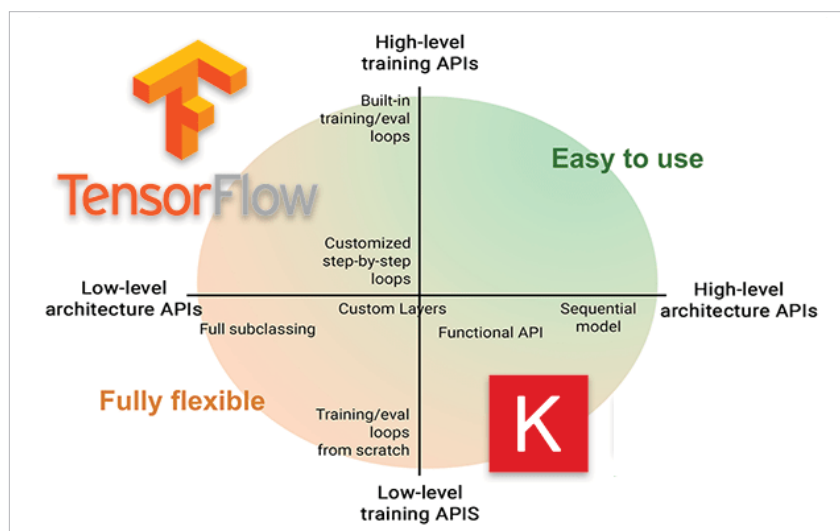


Figure 1: Why TensorFlow is popular for deep learning frameworks

```

# example of loading and plotting the mnist dataset
from tensorflow.keras.datasets.mnist import load_data
from matplotlib import pyplot
# load dataset
(trainX, trainy), (testX, testy) = load_data()
# summarize loaded dataset
print('Train: X=%s, y=%s' % (trainX.shape, trainy.shape))
print('Test: X=%s, y=%s' % (testX.shape, testy.shape))
# plot first few images
for i in range(25):
    # define subplot
    pyplot.subplot(5, 5, i+1)
    # plot raw pixel data
    pyplot.imshow(trainX[i], cmap=pyplot.get_cmap('gray'))
# show the figure
pyplot.show()

```

Figure 2: TensorFlow Python code to implement CNN deep learning model

Table 1: TensorFlow vs PyTorch vs Keras

Criteria	TensorFlow	PyTorch	Keras
Developed by	Google	Meta (Facebook)	Google
Written in	C++, Python, CUDA	Lua	Python
Compatibility	Large data set	Large data set	Small data set
Debugging capability	Difficult	Good	Not required
Choice of use	Scalability and performance	Flexibility and dynamic nature	Ease of use and simplicity
Popular industry use cases	Image classification, language translation, speech recognition, etc	Computer vision, NLP, recommendation system, etc	Speech recognition, healthcare, sentiment analysis, text classification

using the following command:

```

import tensorflow as tf
# Creating a tensor
a = tf.constant([5, 7])
print (a)

```

Variables: Parameters of a model, such as a neural network's weights and biases, are stored and updated in variables. They are created using the `tf.Variable()` function, which takes an initial value as its argument. The following command can be used to create variables in TensorFlow:

```

import tensorflow as tf
# Creating a variable
b = tf.Variable(23)
print (b)

```

Placeholders: When training or inferencing a TensorFlow model, placeholders are frequently used to inject input data into the model. You can use the `tf.placeholder()` function to create a placeholder in TensorFlow. This function takes two arguments: the data type of the tensor that will be fed into the placeholder and the shape of the tensor (which can be optional). Here is an example of creating a placeholder for a 2D tensor of floats:

```

import tensorflow as tf
# create a placeholder for a 2D tensor of floats
x = tf.placeholder(tf.float32,
shape=(None, 2))

```